

Kitchen Kompanion App

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Development Platform: HTML/CSS/Javascript

Instructions

The initial html file is index.html. To launch the app, open the index.html in the unzipped file. The app is fully functional upon launch and there is no additional setup needed.

Upon launch of the app, the app defaults to the recipe book page. There is a navigation bar at the bottom of the screen for users to navigate between pages. This app has 5 pages - recipe book, kitchen contents, meal planning, grocery list, and profile. To navigate to a new page, simply click on any of the icons in the navigation bar.

Please refer to individual tasks in the task list and vertical design write-ups for detailed guides on how to use the tab page.

Task List

Deep Tasks

Deep Task 1: Discover and Follow and Personalized Recipe

- Navigate to the Recipe Tab - Initial Action
 - The user navigates to the Recipes tab by clicking the corresponding button at the bottom of the app.
 - The app displays a feed of recipe cards
- Select a Recipe - Dropdown Selection
 - The user selects a recipe card to open the Recipe Detail view
 - The screen will show ingredients, preparation time, and step-by-step instructions
 - An ingredient will be red if it is missing or substitution required
- Ingredient Substitution – Customization
 - Ingredient substitution is enacted when the user is missing an ingredient or needs substitution for dietary restrictions purposes.
 - If the user lacks or is allergic to an ingredient, the app suggests substitutions (e.g., oat milk instead of dairy milk).
 - Substitution will be based on the user's inventory and what they already have
 - User can accept, reject or manually edit the alternative
- Follow the Recipe – Step-by-Step Interaction
 - The user enters Cooking view, which displays instructions one step at a time
 - After completing all steps, the app prompts the user to confirm that the recipe was followed successfully
 - Upon confirmation, the app automatically updates inventory with all the ingredients used

Deep Task 2: Creating/Using a Grocery List Template

- Tapping the New List (+) button - Initial Action
 - To create a template, the user will click on the New List button to bring up the page to create a new list
- Add Items to the List - Customization/Data Entry
 - Add items to the list as normal
- Create a Template - Required Action
 - At the bottom of the page, there will be two buttons: Add to Templates or Create New List
 - User clicks on the Add to Templates button
- Using a Template - Optional Action
 - To use a template, user clicks on the Templates button next to the New List button to bring up the Template Library
 - Click "Add" on a template to add it to current lists

Deep Task 3: Updating Kitchen Contents

- **Navigate to Kitchen Contents Tab - Initial Action**
 - the kitchen contents tab is one of the main tabs at the bottom of the screen
- **Select Location - Filtering**
 - At the top of the main kitchen contents, there will be three filters: All, Fridge, and Pantry to filter the contents by location in the kitchen
- **Add Items to Inventory - Customization/Data Entry**
 - if not already on the list: add the name of item including its quantity, update as available, and add the location
 - if already on the list: increase the quantity and note the different expiration dates

Deep Task 4: Create and Manage a Weekly Meal Plan

- **Navigate to the Meal Planning Tab – Initial Action**
 - The user taps the Meal Prep icon in the bottom navigation bar to open the Meal Planning tab.
 - The tab displays the current week (e.g., Week of Nov 3 – Nov 9) with separate sections for Breakfast, Lunch, and Dinner.
 - Each slot starts with a “+ Add meal” button where users can begin planning their meals.
- **Add or Edit Meals – Customization/Data Entry**
 - When the user taps “+ Add meal,” a pop-up form appears with fields for meal name, servings, prep time, and optional notes.
 - After saving, the meal appears in the selected day’s section, showing the name, servings, and prep time.
 - Users can also edit or remove meals using the Edit and Remove buttons next to each entry.
- **Generate or Clear a Plan – Optional Action**
 - The user can click “Generate Plan” at the top of the screen to automatically create a weekly plan based on their saved recipes or preferences.
 - The Clear button resets the current plan so users can start over.
 - The Export option allows users to save or share their plan.
- **View and Navigate Weeks – System Feedback**
 - The header displays the current week, and users can scroll between past or upcoming weeks (e.g., Week of Dec 8 – Dec 14) using the arrow buttons.
 - Notes under each recipe, such as “no mushrooms” or “add avocado if available,” help users personalize their plan.

Moderate Tasks

Moderate Task 1: Rate and Save a Recipe

- Provide Rating and Feedback – Data Entry
 - The user selects a star rating (1–5) to evaluate the recipe.
- Save to Favorites – Optional Action
 - The app stores the feedback and updates personalization algorithms to improve future

Moderate Task 2: Adding a Grocery List Item

- Add a New Item - Initial Action/Data Entry
 - User can add a new item using the + button in the list header to add it to the list
- Add More Information - Data Entry/Optional Action
 - The + button brings up a modal that allows the user to add the name of an item, quantity, and price
 - Each item has a pencil icon to edit item information, including name, quantity, and price

Moderate Task 3: Add or Swap a Meal in the Weekly Plan

- Select a Meal Slot – Initial Action
 - The user taps on any empty meal slot in the weekly layout (e.g., Wednesday Dinner).
 - A pop-up form appears labeled “Add Meal” with input fields for meal name, servings, prep time, and optional notes.
- Enter Meal Details – Data Entry Step
 - The user types in the information for their meal (for example, “Chicken Fried Rice,” 4 servings, 35 min prep time, notes: no mushrooms).
 - After filling out the fields, the user clicks “Save Meal.”
 - The meal now appears in that time slot with its details displayed.
- Edit or Remove a Meal – Optional Action
 - If the user wants to make changes, they can tap “Edit” next to an existing meal to update its details or “Remove” to delete it.
 - After editing, the system updates the displayed information instantly.

Moderate Task 4: Customize Profile Preferences

- Fill in profile information - Initial Action/Data Entry
 - Users can scroll through the profile page and input information specific to them, such as name, roommate information, and dietary/cuisine preferences
- Save Preferences - User Action
 - Users can click the Save Profile button to save their preferences
- Edit Preferences - Optional Action

- When the user returns to the Profile page, their preferences will be automatically saved
- The user can change any of the information and click the Save Profile button again to save new changes

Shallow Tasks

Shallow Task 1: Checking off Grocery List Items

- Check off Item - User Action
 - Users can click the item to cross it off the list

Shallow Task 2: Deleting Grocery List Items

- Delete Item - User Action
 - Users can click the trash icon next to the item to delete it from the list
- Delete List - User Action
 - Users can click the trash icon next to a list to delete the entire list

Shallow Task 3: Filter Kitchen Items by Availability

- Filter by in stock - User Action
 - Users can click on a filter that automatically shows if a kitchen item is available or needs to be bought

Shallow Task 4: Clear a Meal Slot

- Locate the Meal Slot – Initial Action
 - The user taps the “...” icon beside any planned meal (e.g., “Friday Lunch”).
- Clear the Slot – Single Action
 - From the dropdown, the user selects “Clear Meal.”
 - The recipe name disappears, and the cell resets to display “+ Add Meal.”
 - A quick confirmation message appears: “Meal removed.”

Shallow Task 5: Clear Profile Preferences

- Clear profile - Initial Action
 - Users can click on the Reset Profile button in the Profile page to clear their profile preferences from local storage

Horizontal Design

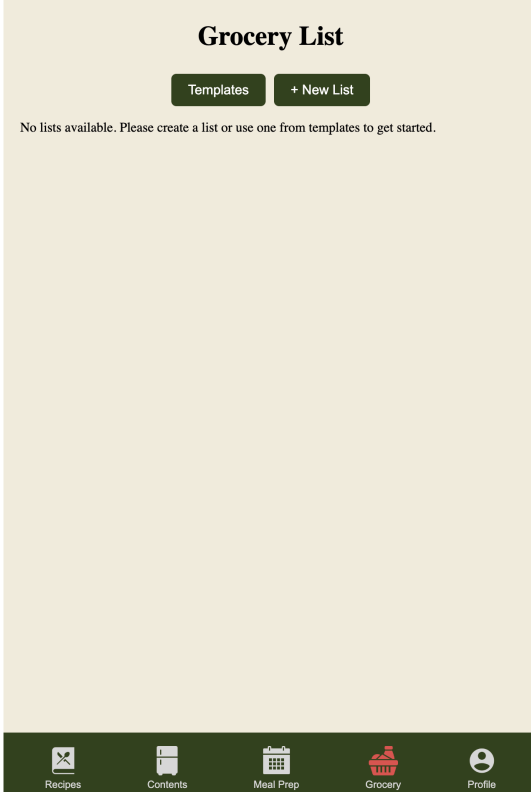
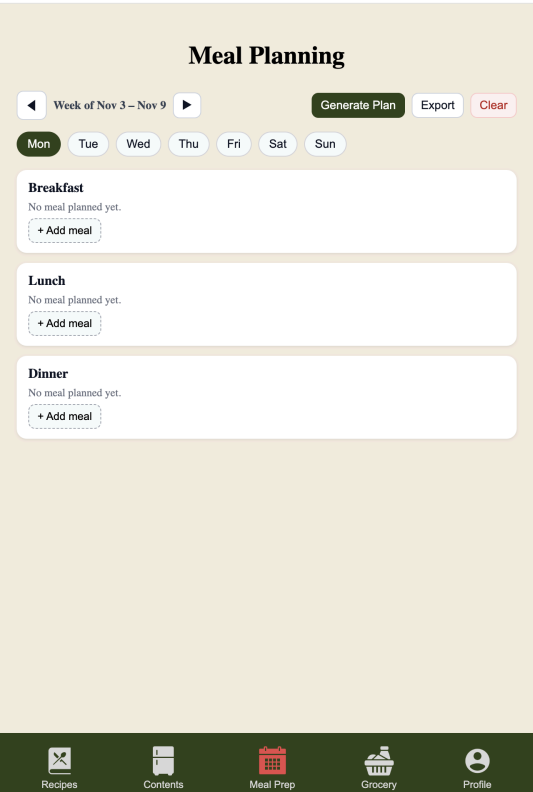
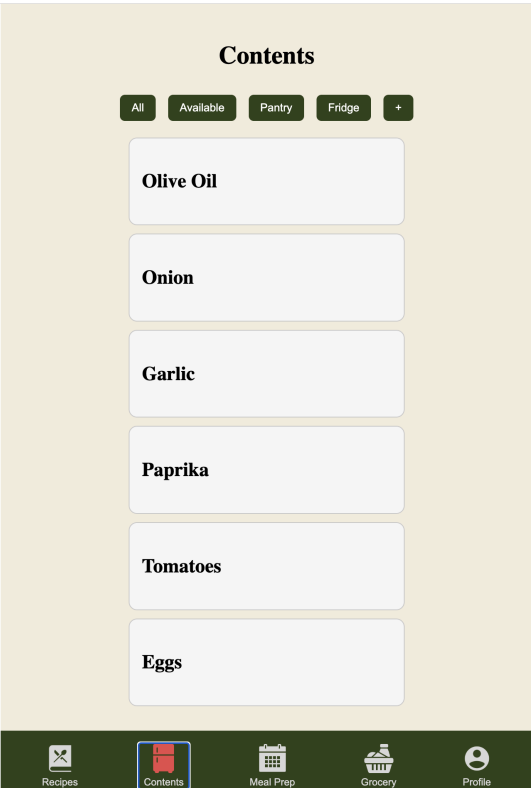
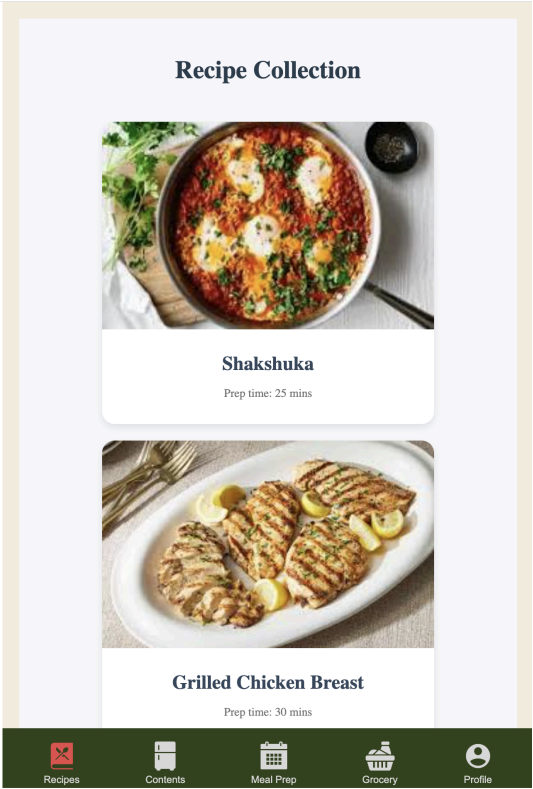
The Kitchen Companion app is designed to help users efficiently manage their cooking and meal planning experience, as well as activities associated with the kitchen in one cohesive platform. The goal of the app is to make everyday kitchen tasks such as finding recipes, tracking inventory, planning meals, and organizing grocery lists simple, fast, and enjoyable. We focused on creating an interface that is intuitive and accessible for users of all skill levels, from college students, beginner home cooks, to experienced meal preppers.

Each screen in our app follows consistent visual language through usage of colors, typography, imagery, and layout structure. We used a soft dark green and white to represent freshness and align with the app's cooking theme and maintain strong contrast for readability. The Times New Roman font throughout the app ensures clear legibility and a modern feel. The buttons, icons, and input components of the app share uniform shapes and spacing across all pages, reinforcing internal consistency so users can easily predict how to interact with new elements as they navigate different tabs.

On the horizontal level, we have a fixed navigation bar at the bottom of the screen that provides quick access to the app's five main sections. Each tab includes an icon and a text label for recognition. The active tab icon also changes color to signal the user which tab they are currently on. This form of visibility of system status reduces confusion. The icons also correspond to common real life scenarios (such as a book for recipe selection and a basket for groceries). We also took advantage of the positive transfer effect by using familiar symbols from other cooking and productivity apps. This external consistency minimizes the learning difficulties for new users.

Our layout follows a clear English reading flow, with information organized from top to bottom and left to right. Important elements like titles and buttons are placed near the top, while the main content sits in the middle and the navigation bar stays fixed at the bottom for easy thumb reach. This structure makes the app simple to scan and understand at a glance. We also used spacing and grouping to keep the screens clean and organized, reducing clutter and helping users focus on one area at a time. Accessibility was another focus and text sizes were increased where needed, touch targets were made larger, and icons were paired with text labels to make sure all users can easily recognize actions.

Top Level Screenshots



Profile

Name:

Jane Doe

Sharing Fridge?

- ☒ Yes
- ☐ No

Number of roommates:

3

Dietary restrictions:

- ☒ Vegetarian
- ☐ Vegan
- ☐ Gluten-Free
- ☐ None
- ☐ Other



Recipes



Contents



Meal Prep



Grocery



Profile

Vertical Design

Recipe Page



Shakshuka

Prep time: 25 mins

Ingredients

- 2 tbsp olive oil
- 1 onion, chopped
- 1 red bell pepper, diced
- 3 cloves garlic, minced
- 1 tsp paprika
- 1 tsp cumin (missing)
- 1 can (14 oz) crushed tomatoes
- 4 large eggs
- Salt and pepper to taste
- Fresh parsley for garnish

[Add Missing Ingredient to Grocery List](#)[Start Cooking](#)

[← Back to Recipes](#)

Cooking Shakshuka

Ready to cook? Click "Next Step" to begin.

[Next Step](#)[View All Steps](#)

[← Back to Recipe](#)

Cooking Shakshuka

1. Heat olive oil in a large skillet over medium heat.

[Next Step](#)[View All Steps](#)

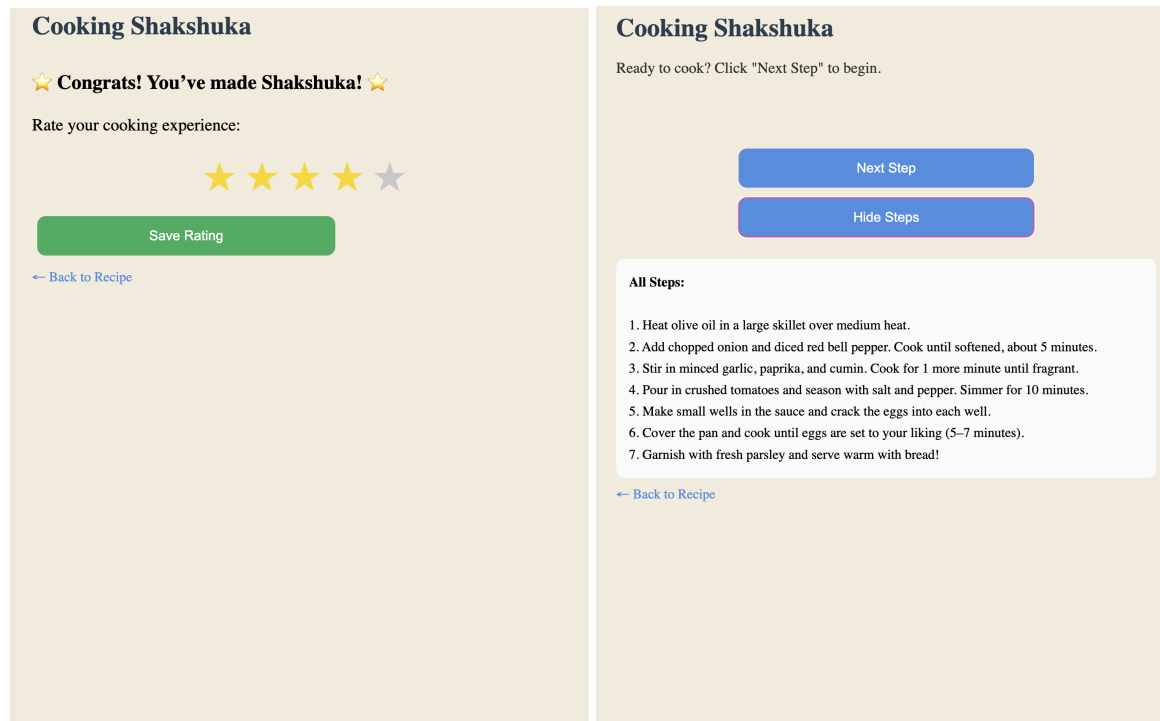
[← Back to Recipe](#)

Cooking Shakshuka

2. Add chopped onion and diced red bell pepper. Cook until softened, about 5 minutes.

[Next Step](#)[Previous Step](#)[View All Steps](#)

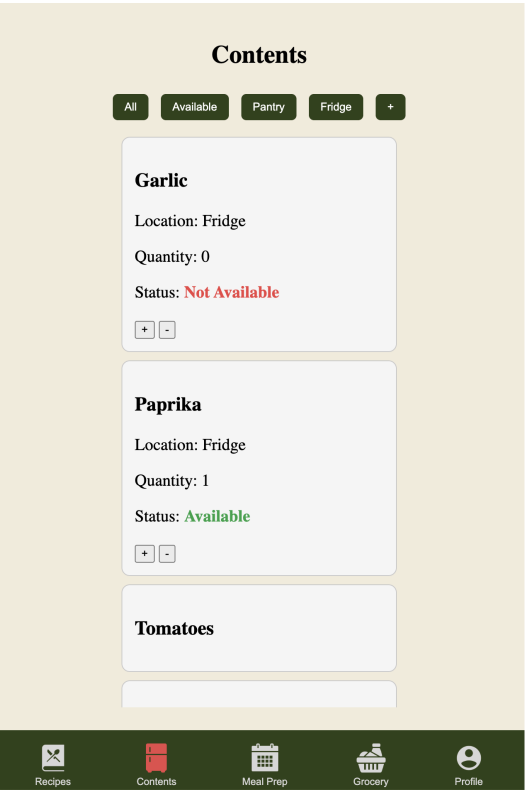
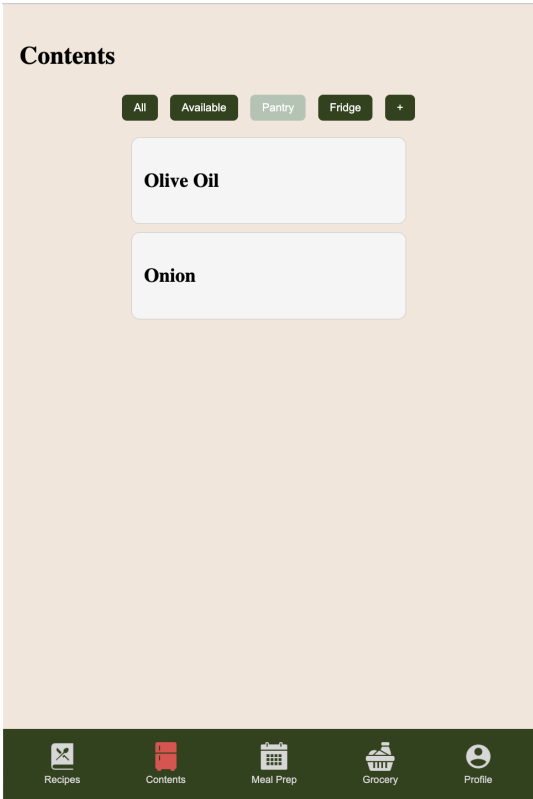
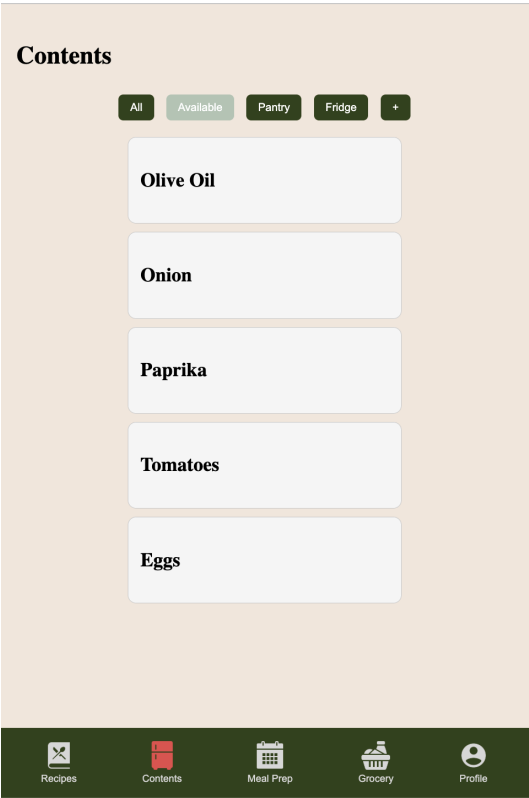
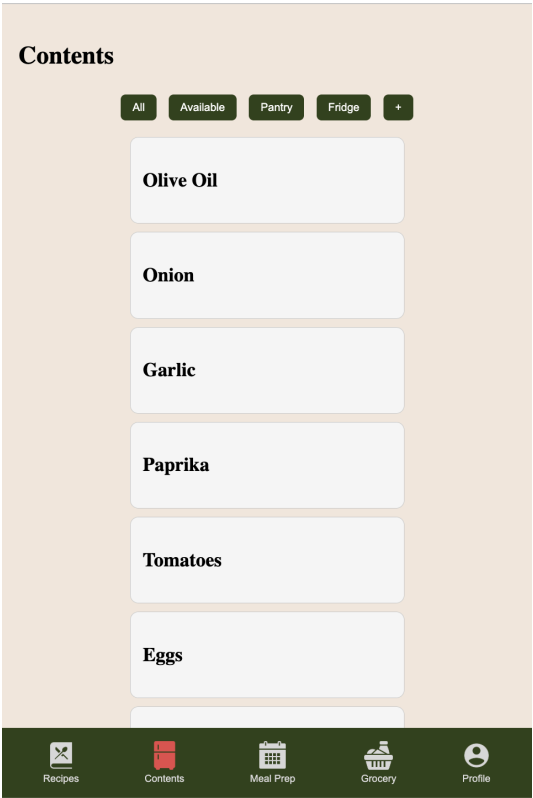
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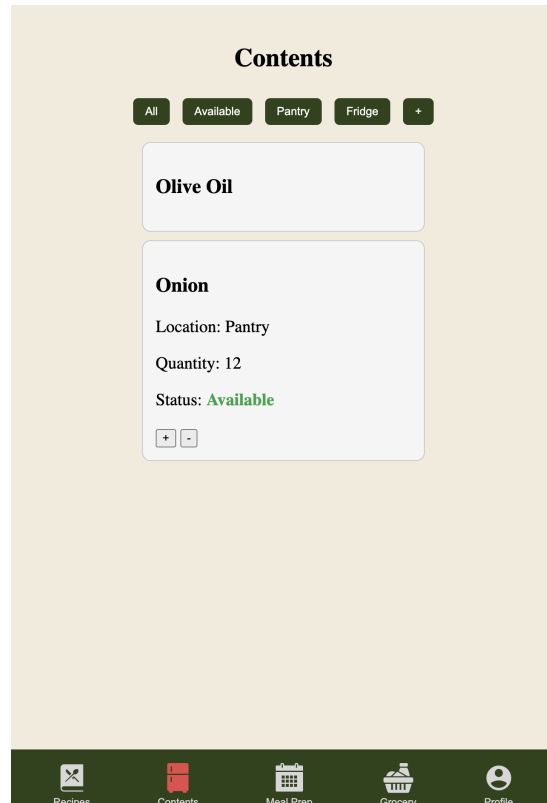
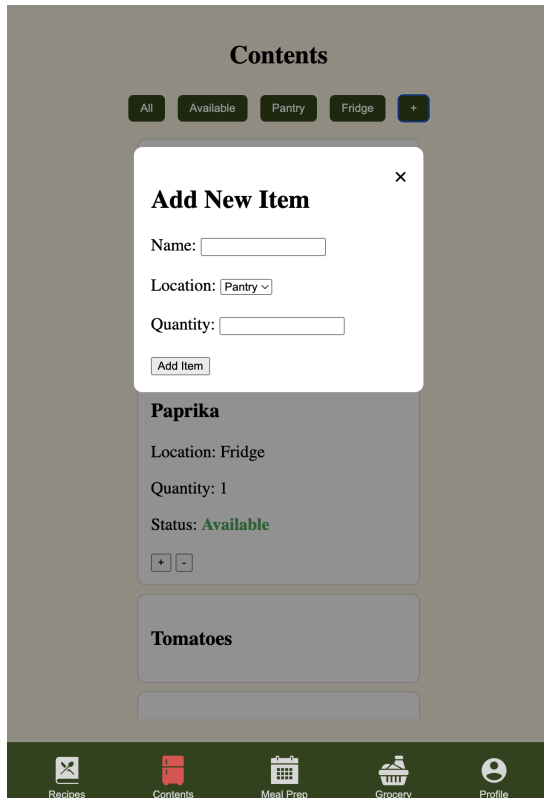


One of our vertical tasks is the Recipe Book page. When the user clicks into the tab, they have a list of recipe cards to pick from. They can scroll through the page to pick a recipe they want. Let's say they pick the first recipe, Shakshula for example. They click into it and they will see the ingredients, prep time, and a bigger picture of the food. The ingredients are also color coded. If it's red it means the user is missing that ingredient in their pantry. However, if the start cooking button is present, that is an indication that the ingredient is not essential to the recipe and it can be optional if really needed. So the user can still make the food.

Users also have the option to add the missing ingredient to their shopping list. This will be done automatically. Although this feature is not yet implemented. When the user clicks start cooking, they are led to a cooking view where each step of the cooking is displayed on a screen with the previous step button and next step button. This step-by-step presentation reduces cognitive overload by focusing the user's attention and preventing mistakes. The screen also has an option to display all steps at once, as well as hiding those steps. Once the user clicks through all the steps, there is a rating option where the user can rank the recipe, cooking experience by a 5 star system. This serves as positive reinforcement, motivates engagement, and provides implicit feedback to the system. In the future, these ratings, combined with pantry data and user preferences, will allow the system to recommend recipes tailored to the user's tastes and available ingredients, following a user-centered design approach. Additional improvements include integrating the "Add Missing Ingredient" button with the grocery list page to automatically update grocery lists with missing ingredients.

Kitchen Contents





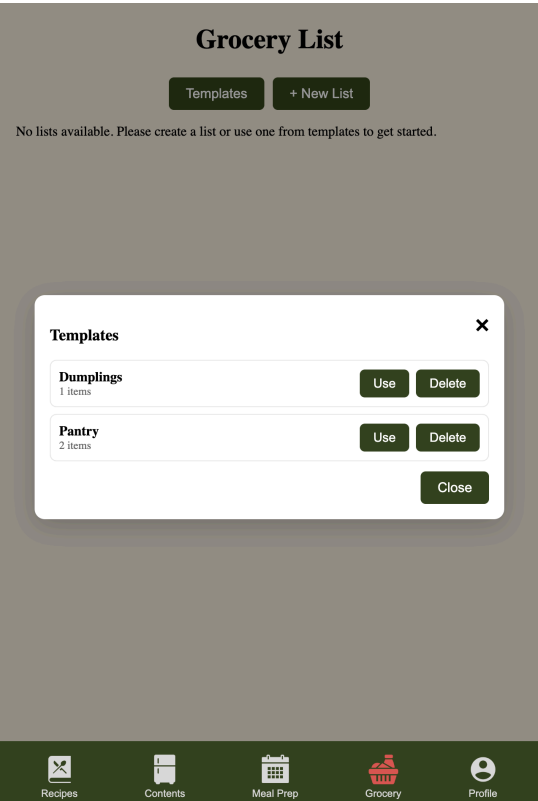
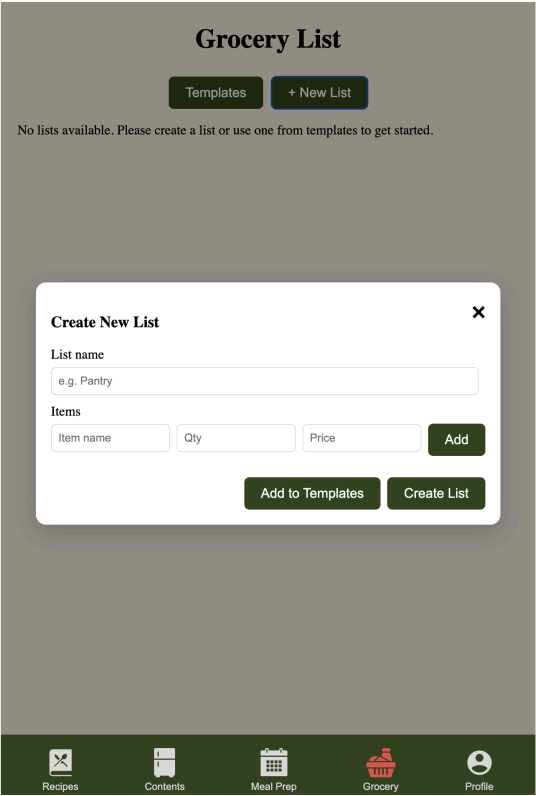
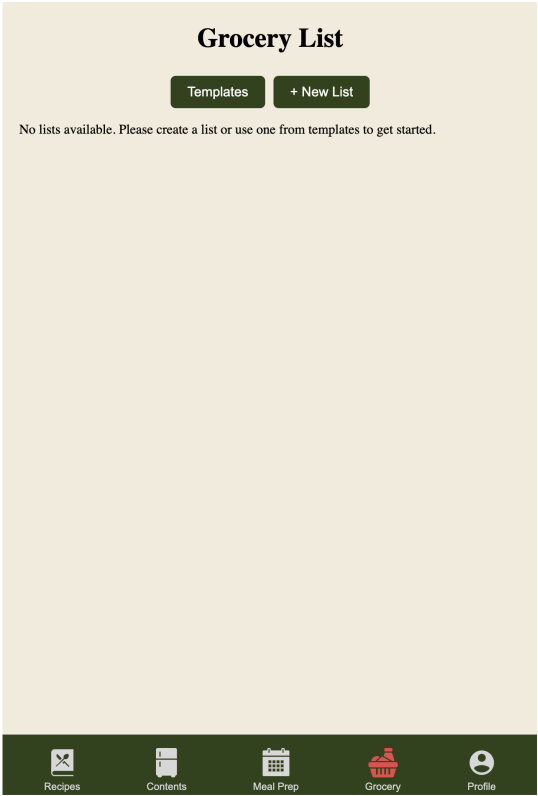
The Kitchen Contents Page allows the user to sort their items based on availability as well as its location. The initial page shows all of the items that they may have stored. In order to see the items that are available, the user could click the available button that then only shows the items that are available. In addition, the user could sort the items based on their location in the kitchen. The items found in the pantry are shown if the pantry button is clicked and the items found in the fridge are shown if the fridge button is clicked.

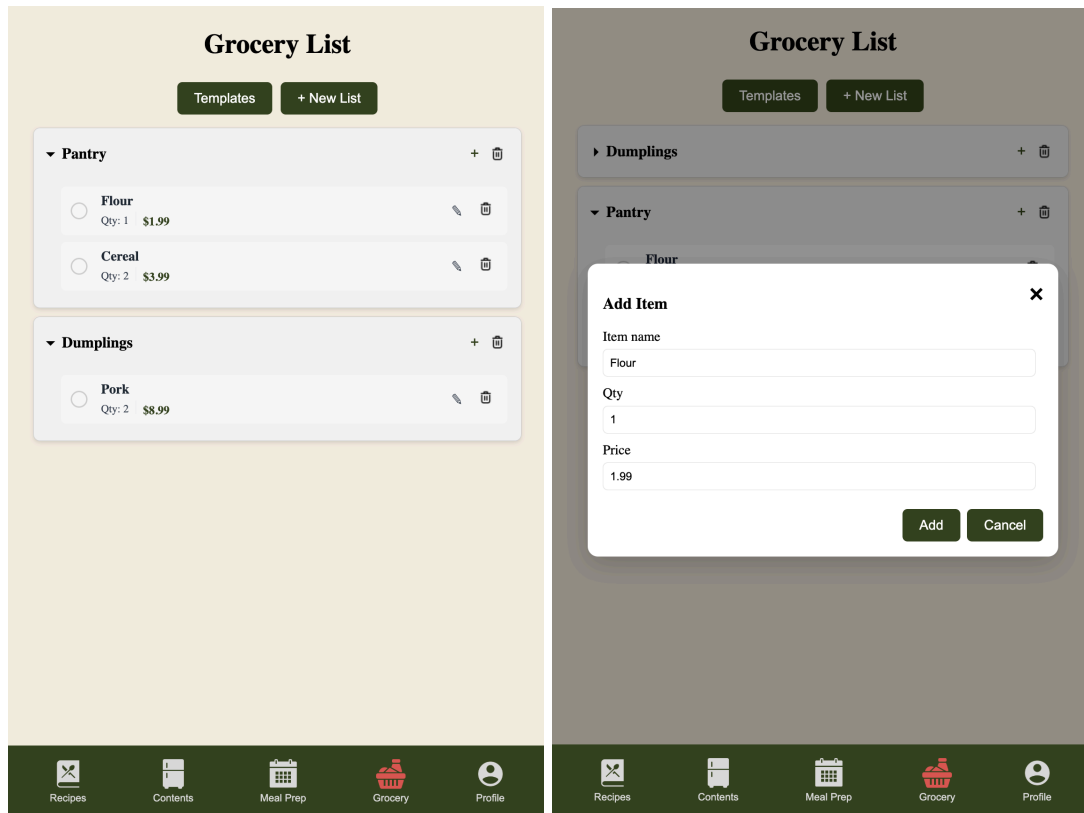
Users also have the ability to add items to their kitchen contents inventory. In order to do so, the user can press the + button shown at the top of the page next to the sort buttons. A pop up then appears. The user is prompted to enter the name of the item, where its location in the kitchen is, and the quantity of the item being added. After the information is filled out, the user can click add item and the item is added to the kitchen contents tab.

Integrating an add item button at the top of the page makes it easier for the user to be able to use this feature. Utilizing the '+' sign allows for a transfer effect so that the user knows that it is used to add something. Further, the initial design had a pop-up that showed the details of an item, but I redesigned it so that it acts as a drop-down instead. This allows the user to see the details of more than one ingredient at a time. In addition, I believe that sorting based on availability works well, as well as filtering out the items based on where it is located. However, I think this could be

improved by allowing the user to be able to filter out the items based on both location and availability and not just one or the other. Further, integrating this page to the other pages to automatically update the quantity of an item could be improved.

Grocery List





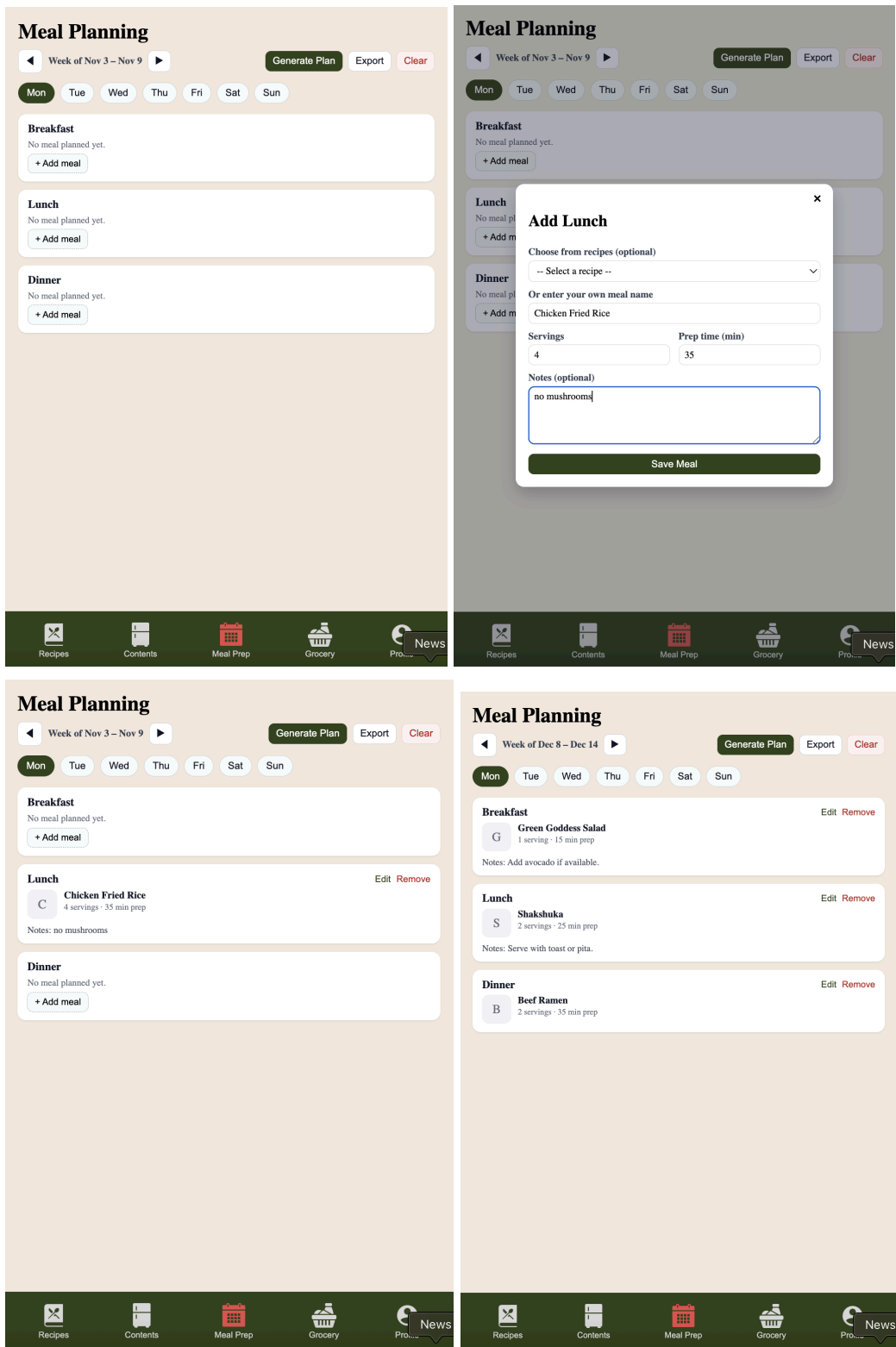
The design for the grocery list page includes in-depth functionality for basic grocery shopping features, such as creating lists and adding and crossing off items on a list. However, to support multiple task scenarios, there are additional features such as adding quantity and price to items, creating multiple lists, and saving lists to templates. To integrate these features seamlessly, the grocery list tab has two buttons, a “Templates” button and “New List” button. The user can click on the “New List” button to bring up a “Create New List” modal, which lets the user input a list name, and add any starting items, quantity, and price to the list. They can then confirm the list and click on the “Create List” button to add it to the main page. The modal automatically closes, and the user can see the new list on the main page. This list is expandable to see all items, and users can click the + button in the header to add additional items, which brings up the “Add Item” Modal where users can once again input item information. Users can also edit any item using the pencil icon to the right of the item. The trash icon is present next to each individual item and list to delete the element. Users can also tap on the circle check to the left of each item to cross it off the list. Using standard icons, like trash for delete and pencil for edit, uses the transfer effect and cultural standards to bring familiarity to the user. They can easily pick up on common actions like add, delete, and edit based on familiar icons.

To create a template, the user creates a new list as usual from the “New List” button, but instead of clicking the “Create List” button, the user clicks on the “Add to Template” button. This adds a template to the template library, which users can select from to add any template to the main

page. To access the template library, users can click on the “Templates” button on the main page, and click “Use” or “Delete” to use or remove the template. This reduces the amount of time spent creating identical or similar lists, and increases user efficiency when using the feature.

The current functionality of these tasks works well, and is easily understandable for the user. Parts that need improvement include consistent styling across pages, as well as integration between the recipe and kitchen contents pages. If time permits, adding in a sharing function for grocery lists will also be implemented.

Meal Planning



The Meal Planning tab helps users plan their meals for the week in a clear and organized layout. At the top, users can view the current week, switch between different weeks, and use buttons like “Generate Plan,” “Export,” or “Clear” to manage their schedule. The layout shows each day of the week as a button, with sections for Breakfast, Lunch, and Dinner underneath. Each section includes a “+ Add meal” button that opens a pop-up form where users can enter details such as meal name, servings, prep time, and notes. Once saved, the meal appears with its details and options to Edit or Remove, allowing users to easily make changes if their plans shift.

The “Generate Plan” feature automatically fills in meals for the week based on saved recipes or preferences, which helps users who want quick, ready made suggestions. The “Export” button is designed to make meal prep more practical by allowing users to save or share their plan for the week. Users can also view different weeks (like Week of Nov 3 – Nov 9 or Week of Dec 8 – Dec 14) to track what they have planned or reuse past meals and recipes. The interface keeps everything visually simple by using clear spacing, consistent buttons, and a logical flow for the user that makes it easy to navigate.

Overall, this layout supports both flexibility and structure for users. It works well for users who want to manually enter their meals as well as those who prefer auto generated plans. In the future, I would like to make the page more interactive by adding drag-and-drop features and small icons or color cues to show completed days or recipes that are ready to cook based on the ingredients the user has in their pantry.

Profile Page

Profile

Name:

Sharing Fridge?

☒ Yes

☐ No

Number of roommates:

Dietary restrictions:

☒ Vegetarian

☐ Vegan

☐ Gluten-Free

☐ None

☐ Other

☐ Vegetarian

☐ Vegan

☐ Gluten-Free

☐ None

☐ Other

Cuisine preferences:

☐ Italian

☐ Chinese

☐ Mexican

☐ Indian

☐ American

☐ Other

[Reset Profile](#) [Save Profile](#)

 Recipes

 Contents

 Meal Prep

 Grocery

 Profile

 Recipes

 Contents

 Meal Prep

 Grocery

 Profile

The profile page allows users to save personal preferences, which can be used to personalize recipes. It comes with a couple of options, such as inputting your name, if you're sharing the fridge, number of roommates, dietary restrictions, and cuisine preferences. The user has the option to input any fields they want, and when they scroll to the bottom, there will be a button that allows the user to save their profile. Once saved, the preferences will be saved to local storage, so the user can access them any time. If the user wishes to edit their profile, they can edit any field and click the "Save Profile" button once again to save their preferences. The user can also reset their profile and clear all preferences.